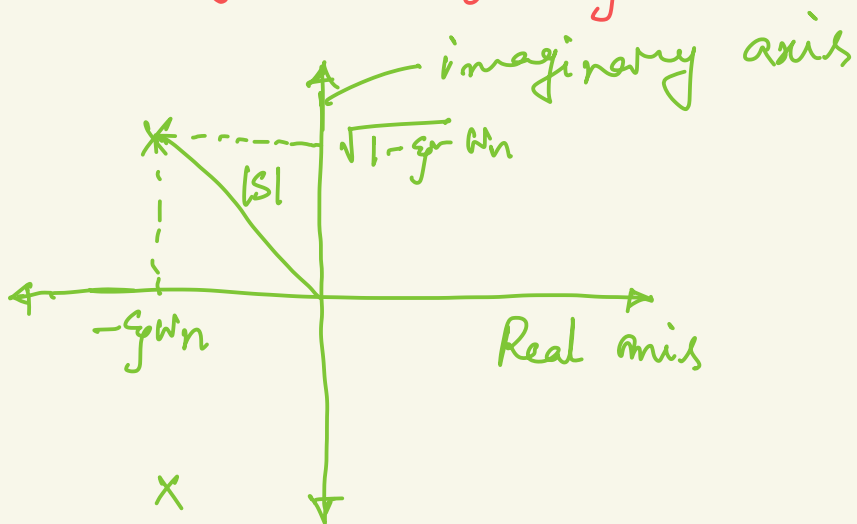




Lecture 16

In this lecture, we would like to investigate some geometric properties of the poles of the second order transfer functions. The poles are

$$s_{1,2} = -\zeta \omega_n \pm j \sqrt{1-\zeta^2} \omega_n$$

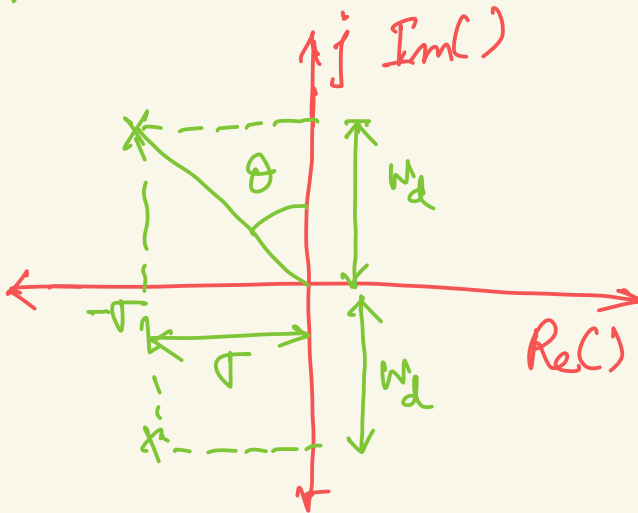


$$\begin{aligned} |s|^2 &= (-\zeta \omega_n)^2 + (\sqrt{1-\zeta^2} \omega_n)^2 \\ &= \zeta^2 \omega_n^2 + (1-\zeta^2) \omega_n^2 \end{aligned}$$

$$= \xi \tilde{\omega}_n + \tilde{\omega}_n - \xi \tilde{\omega}_n$$

$$|S| = \tilde{\omega}_n$$

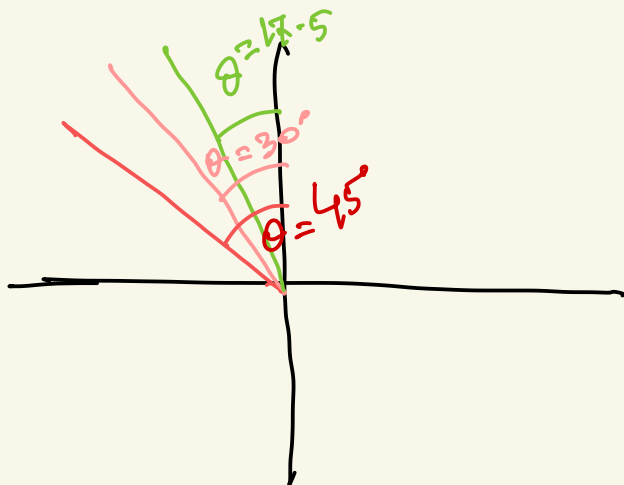
$$|S| = \omega_n$$



$$\sin \theta = \frac{\sigma}{|S|} = \frac{\sigma}{\omega_n} = \frac{\xi \omega_n}{\omega_n}$$

$$\sin \theta = \xi$$

$$\theta = \sin^{-1} \xi$$



$$\theta = 45^\circ, \quad \zeta = 0.707$$

$$\theta = 30^\circ, \quad \zeta = 0.5$$

$$\theta = 17.5^\circ, \quad \zeta = 0.3$$

when θ decreases, ζ decreases too.

On the real axis $\theta = 90^\circ, \quad \zeta = 1$

On the imaginary axis $\theta = 0^\circ, \quad \zeta = 0$

A lower value of damping

ratio means more oscillations.

